

Publication Bias in Statistical Meta-Analysis

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Outline

1. Objective
2. Introduction to Statistical Meta-Analysis
3. Dataset
 - a. Context
 - b. Preliminary data exploration
4. Meta-Analysis
 - a. Fixed-effect
 - b. Random-effects
 - c. Meta-regression
 - d. Interpretation
5. Publication Bias
 - a. Introduction
 - b. Graphical and statistical tests
 - c. Interpretation
6. Overall Conclusions
7. References
8. Appendix

**Explore and discuss the
effects of publication bias
in statistical
meta-analysis.**

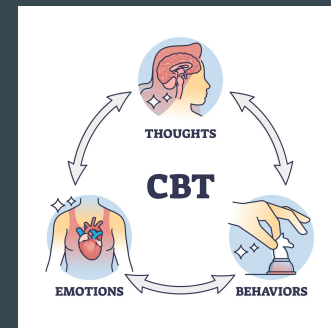
Introduction to Meta-Analysis



- “Analysis of analysis”
 - Resolve conflicting results between a number of independent studies
 - Studies themselves serve as the sample
- Overall conclusion
 - Based on pooled estimate(s) of effect size(s)
 - Primary measure of interest
 - Reported by studies
 - Mean difference, relative risk, correlation coefficient
 - Associated standard error
 - Occasionally multiple (multivariate meta-analysis)
 - Weighted average dependent upon precision of individual studies

Dataset

Collection of studies on effectiveness of cognitive behavioral therapy (CBT) on recidivism. Treatment group received therapy, control group did not.



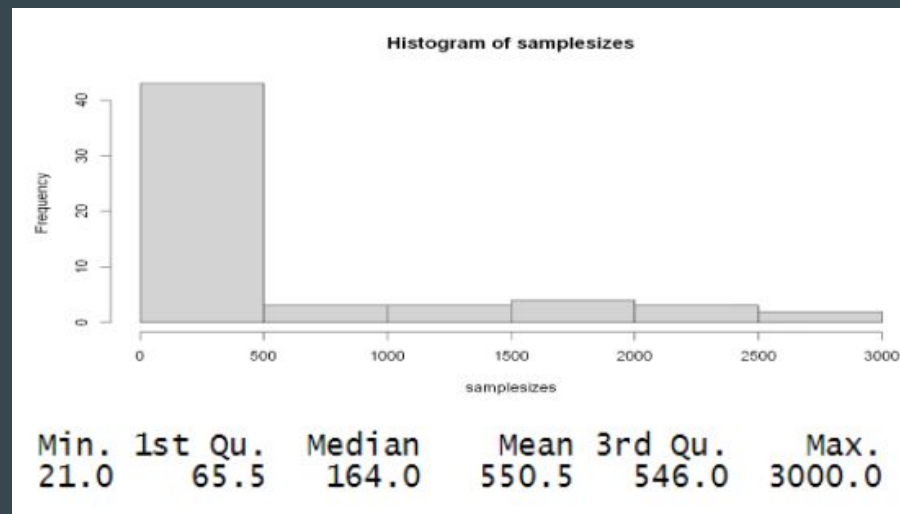
- Number of studies: $k = 58$
- Independent variables: $p = 27$
 - Categorical and continuous
- Effect size: log-transformed odds ratio
 - 4 groups: control recidivists (A), control non-recidivists (B), treatment recidivists (C), treatment non-recidivists (D)

$$OR = \frac{A*D}{B*C}, \log(OR) = \log\left(\frac{A*D}{B*C}\right), SE(\log(OR)) = \sqrt{\frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D}}$$

Preliminary Data Exploration

study <chr>	pubtype <chr>	country <chr>	design <chr>	program <chr>	setting <chr>	designprob <chr>	n.ctrl.rec <int>	n.ctrl.non <int>	n.cbt.rec <int>	n.cbt.non <int>
Guerra (1990)	journal	USA	rct	research	prison	no	23	29	10	19
Leeman (1993)	journal	USA	rct	demo	prison	no	15	22	3	17
Kownacki (1995)	chapter	USA	rct	research	community	no	4	6	2	9
Larson (1989)	chapter	USA	rct	research	community	no	11	2	8	5
Finn (1998)	report	USA	rct	practice	prison	no	22	60	16	66
Goldstein (1989)	chapter	USA	rct	demo	community	no	14	18	5	28

- Most studies relatively small, some extremely large
- Wide range of sample sizes → high variability in study precision → conflicting / insignificant results
- Studies with less precision to carry less weight in analyses
- 6 studies reported negative effect sizes



Fixed-Effect Meta-Analysis

- Assume a common effect between all studies
- Sampling error only source of effect size variability between studies
- Goal: estimate single *true* effect size
- Model:

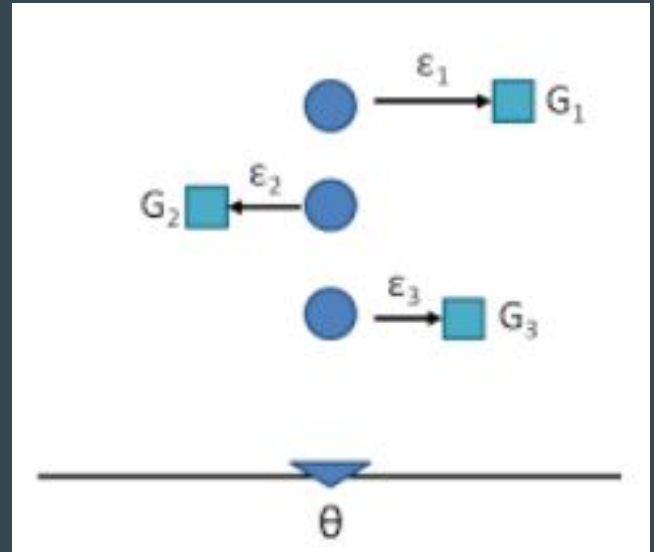
$$\hat{\theta}_i = \theta + \varepsilon_i$$

- Weighted average (overall effect size):

$$M = \frac{\sum_{i=1}^k w_i \hat{\theta}_i}{\sum_{i=1}^k w_i}$$

With inverse-variance weights:

$$w_i = \frac{1}{V_{\hat{\theta}_i}}$$



Fixed-Effects Model (k = 58)

logLik	deviance	AIC	BIC	AICc
-94.6547	213.6898	191.3093	193.3698	191.3808

I² (total heterogeneity / total variability): 73.33%
H² (total variability / sampling variability): 3.75

Test for Heterogeneity:

Q(df = 57) = 213.6898, p-val < .0001

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub	
0.3382	0.0269	12.5870	<.0001	0.2856	0.3909	***

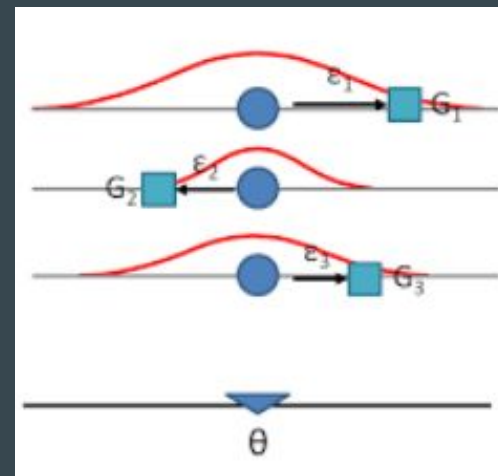
Fixed-Effect Meta-Analysis

Odds Ratio Estimate: 1.402

Random Effects Meta-Analysis

- Two sources of variability (sampling error + heterogeneity)
- No common effect size among all studies
 - Each study has *own* true effect size
- Goal: estimate *mean* of true effect sizes
- Model:

$$\hat{\theta}_i = \mu + \xi_i + \varepsilon_i$$



Weighted
Average:

$$M^* = \frac{\sum_{i=1}^k w_i^* \hat{\theta}_i}{\sum_{i=1}^k w_i^*}$$

Inverse-variance weights:

$$w_i^* = \frac{1}{V_{\hat{\theta}_i}^*}$$

with

$$V_{\hat{\theta}_i}^* = Var(\hat{\theta}_i) + \tau^2$$

Random-Effects Model (k = 58; tau² estimator: DL)

logLik	deviance	AIC	BIC	AICc
-42.5827	109.5459	89.1654	93.2863	89.3836

tau² (estimated amount of total heterogeneity): 0.1196 (SE = 0.0445)

tau (square root of estimated tau² value): 0.3458

I² (total heterogeneity / total variability): 73.33%

H² (total variability / sampling variability): 3.75

Test for Heterogeneity:

Q(df = 57) = 213.6898, p-val < .0001

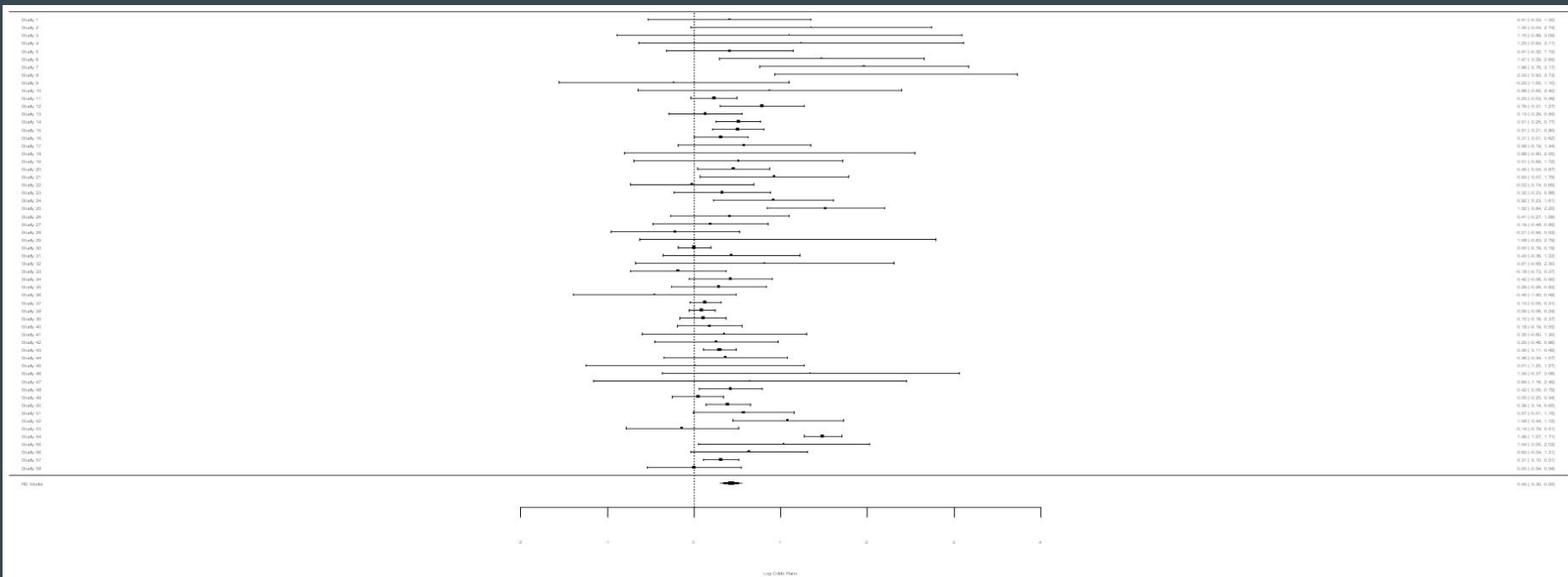
Model Results:

estimate	se	zval	pval	ci.lb	ci.ub	
0.4271	0.0631	6.7633	<.0001	0.3033	0.5508	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Random Effects Meta-Analysis

Odds Ratio Estimate: 1.532



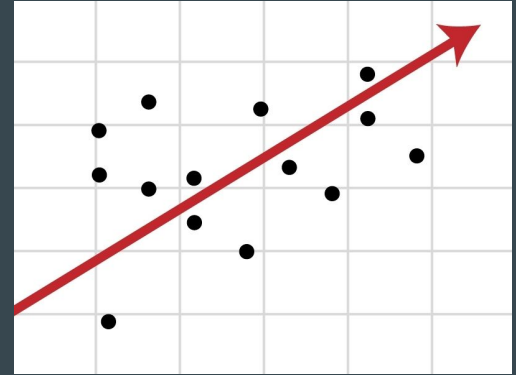
Random Effects Forest Plot

Meta-Regression

- Least Squares Regression:
 - Response / Dependent variable - log-transformed odds ratio (y)
 - Explanatory / Independent variables
 - Average age of individuals (x1)
 - Proportion of males (x2)
 - Total hours of therapy (x3)
 - Use of anger management component (x4)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \varepsilon$$

- Model could be used in prediction
 - Insight as to which variables are *most* effective in rehabilitation
 - Larger parameter estimates suggest a larger reduction in the odds of recidivism



All Moderators

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 43; tau² estimator: REML)

logLik	deviance	AIC	BIC	AICc
-32.6917	65.3833	77.3833	87.2088	80.0930

tau² (estimated amount of residual heterogeneity): 0.1508 (SE = 0.0628)

tau (square root of estimated tau² value): 0.3883

I² (residual heterogeneity / unaccounted variability): 65.24%

H² (unaccounted variability / sampling variability): 2.88

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 38) = 128.6929, p-val < .0001

Test of Moderators (coefficients 2:5):

QM(df = 4) = 1.6094, p-val = 0.8071

	estimate <chr>	se <chr>	zval <chr>	pval <chr>	ci.lb <chr>	ci.ub <chr>
intrcpt	0.6747	0.4600	1.4666	0.1425	-0.2270	1.5764
covarsage	-0.0088	0.0135	-0.6535	0.5135	-0.0353	0.0176
covarsmale	-0.0008	0.0035	-0.2248	0.8221	-0.0077	0.0061
covarstotalhrs	0.0001	0.0004	0.3627	0.7168	-0.0006	0.0008
covarsanger	0.1697	0.1686	1.0065	0.3142	-0.1607	0.5000

Full Meta-Regression Model

CBT Anger Control Component Only

Mixed-Effects Model (k = 58; tau^2 estimator: REML)

logLik	deviance	AIC	BIC	AICc
-40.0462	80.0924	86.0924	92.1685	86.5539

tau^2 (estimated amount of residual heterogeneity): 0.0895 (SE = 0.0319)
tau (square root of estimated tau^2 value): 0.2991
I^2 (residual heterogeneity / unaccounted variability): 66.55%
H^2 (unaccounted variability / sampling variability): 2.99
R^2 (amount of heterogeneity accounted for): 14.49%

Test for Residual Heterogeneity:
QE(df = 56) = 175.9401, p-val < .0001

Test of Moderators (coefficient 2):
QM(df = 1) = 4.5431, p-val = 0.0331

Model Results:

	estimate <chr>	se <chr>	zval <chr>	pval <chr>	ci.lb <chr>	ci.ub <chr>
intrcpt	0.3322	0.0701	4.7366	<.0001	0.1947	0.4696
covars[, 4]	0.2618	0.1228	2.1315	0.0331	0.0211	0.5025

Anger Management Meta-Regression Model

Interpretations and Conclusions

Fixed-Effect

- Extremely significant result
- Significant heterogeneity
- Weighted effect size: 1.402

Random Effects

- Significant result, smaller T.S.
- Accounted for heterogeneity
- Weighted effect size: 1.532

Overall

- CBT significantly reduces recidivism, implying effective rehabilitation
- Apparent overwhelming evidence in favor of conclusion
 - Results of both models significant at confidence levels exceeding 99.999%

Are these results reliable?

Introduction of Publication Bias

- Key assumption
 - Data is comprehensive or representative
- Difficult to include all studies
- Possibility of studies being missed systematically
 - Publication Bias
 - aka Selection Bias
 - Tendency to submit/publish significant studies *only*
 - Publishing negative or insignificant results can hurt the business of therapists



Problems

- Overestimated effect size estimates
- Overlooking negative side-effects
 - Reinforcing invalid beliefs
 - False conclusions

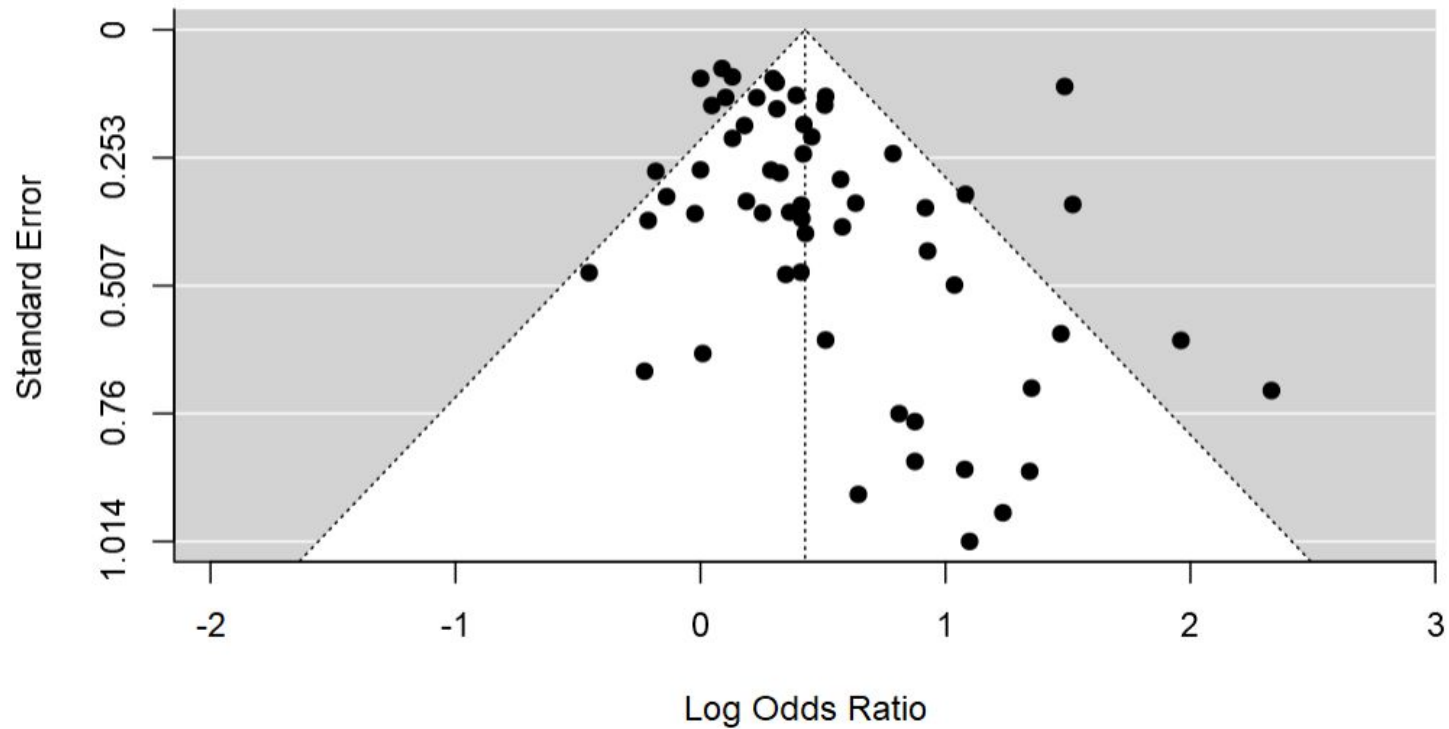


Exploring for Bias (1)

Funnel Plot

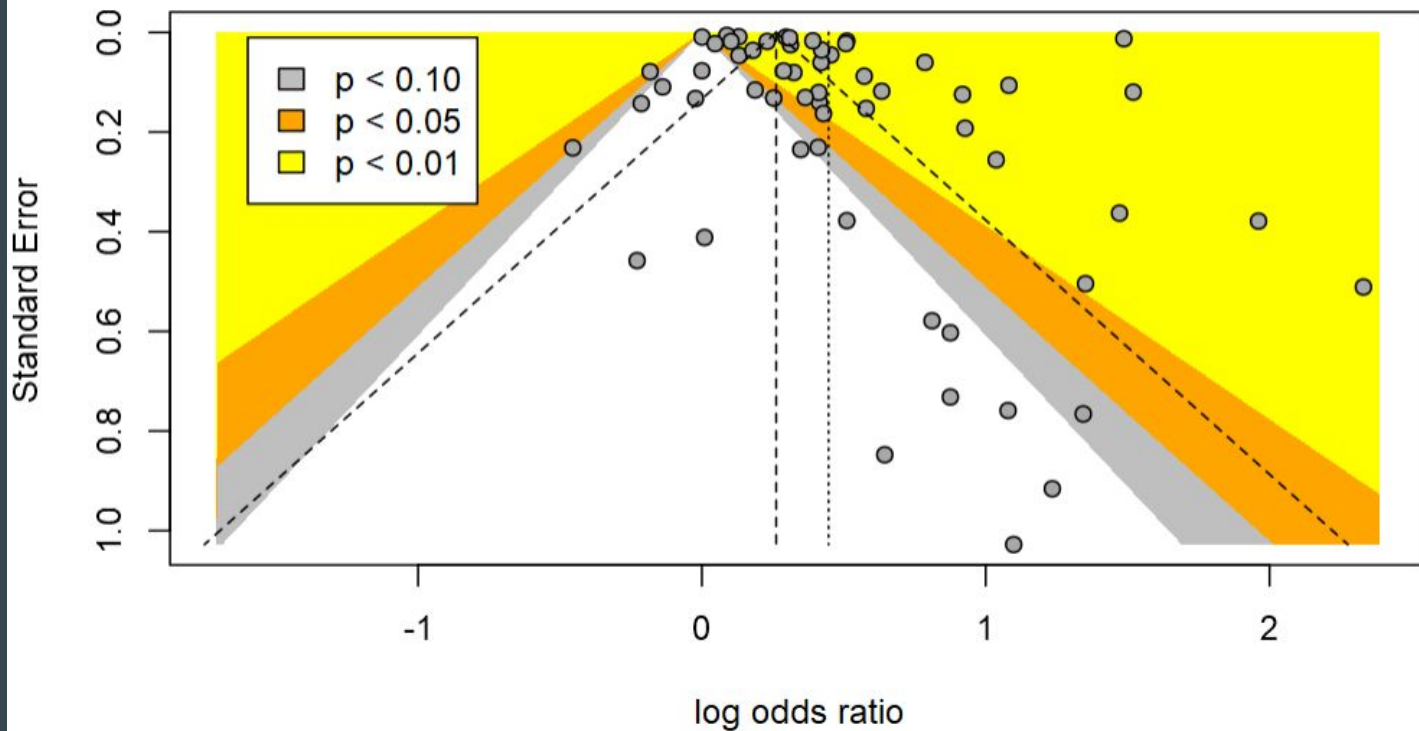
- Plots results from each study
 - x-axis: effect size
 - y-axis: standard error
- Center: overall effect size estimate
 - More precise (i.e. larger) studies → closer to pooled estimate
 - Less precise (i.e. smaller) studies → further from pooled estimate
- Asymmetry suggests publication bias





Funnel Plots

Contour-Enhanced Funnel Plot: Study Significance



Contour Funnel Plot

Exploring for Bias (2)

Egger's Regression Test

- Mathematical test for asymmetry
- Fits the regression model

$$\frac{\widehat{\theta}_k}{SE(\widehat{\theta}_k)} = \beta_0 + \beta_1 \left(\frac{1}{SE(\widehat{\theta}_k)} \right)$$

- Examine the parameter estimate for the slope: β_0
 - Intercept: effect size when precision is 0
 - Precision close to zero with little publication bias
 - Overrepresented small studies → intercept significantly different to 0

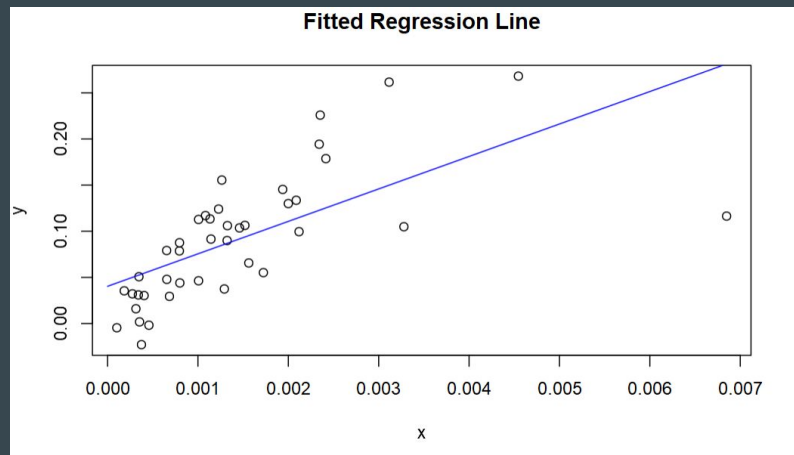


```
Call:
lm(formula = y ~ x, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.164902 -0.029708  0.003049  0.033129  0.111718

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.04036    0.01223   3.301  0.00207 **
x           35.18325    6.39764   5.499 2.57e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05191 on 39 degrees of freedom
Multiple R-squared:  0.4368,    Adjusted R-squared:  0.4223 
F-statistic: 30.24 on 1 and 39 DF,  p-value: 2.565e-06
```



Egger's Regression Test

Addressing Publication Bias

Trim-and-Fill Method

1. Trim

- a. Outlying studies identified and removed
- b. Overall effect size recalculated
- c. New estimate used as “mirror” in center of funnel plot

2. Fill

- a. Hypothetical studies imputed based new center
- b. Imputed studies correspond directly to previously-removed outlying studies
- c. Final pooled effect calculated with *all* original studies and imputed studies



```

Estimated number of missing studies on the left side: 16 (SE = 4.9484)

Random-Effects Model (k = 74; tau^2 estimator: DL)

    logLik  deviance      AIC      BIC     AICc
-78.0518  169.6780  160.1036  164.7117  160.2726

tau^2 (estimated amount of total heterogeneity): 0.2320 (SE = 0.0767)
tau (square root of estimated tau^2 value):      0.4817
I^2 (total heterogeneity / total variability):    82.11%
H^2 (total variability / sampling variability):    5.59

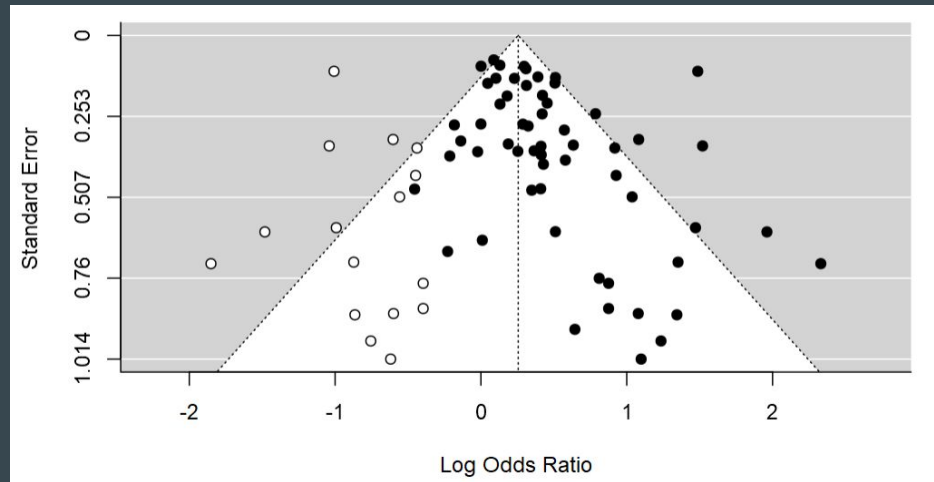
Test for Heterogeneity:
Q(df = 73) = 407.9530, p-val < .0001

Model Results:

estimate      se    zval    pval    ci.lb    ci.ub
  0.2557  0.0730  3.5030  0.0005  0.1126  0.3988 ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```



Trim-and-Fill Results

File-Drawer Problem

- Large effect sizes more likely to be published than smaller ones
- Common in psychological / behavioral analyses
- Inverse normal to combine p-values

$$Z = \frac{1}{\sqrt{k}} \sum_{i=1}^k \Phi^{-1}(P_i)$$

- Fail-safe N: number of insignificant studies needed to change overall conclusion from significant to insignificant

$$k_0 > -k + \left(\frac{\sum_{i=1}^k z_i}{z_1 - \alpha} \right)^2$$

Fail-safe N Calculation Using the Rosenthal Approach

Observed Significance Level: <.0001

Target Significance Level: 0.05

Fail-safe N: 3249



Interpretations

- Funnel plot:
 - Positive effect size asymmetry
- Egger's regression:
 - Significant intercept estimate → funnel plot asymmetry
- Trim-and-Fill:
 - Conclusion *unchanged* despite several insignificant studies imputed
- Fail-safe N:
 - *Many* insignificant studies required to change conclusion



Negative effect sizes are the outliers

Overall Summary

- Overwhelming evidence in support of CBT reducing recidivism
 - Unchanged conclusion, large fail-safe N
 - Not a significant amount of publication bias in dataset
 - **Not known if not explored**
- Crucial to take **ALL** information into account
 - Avoid inaccurate conclusions, strengthen accurate ones
- Failure to explore bias
 - Best case: sets back scientific understanding
 - Worst case: negatively affects *people's lives*
 - Potentially invalidates findings



References

- Landenberger, N. A., & Lipsey, M. W. (2005). The positive effects of cognitive-behavioral programs for offenders: A meta-analysis of factors associated with effective treatment. *Journal of Experimental Criminology*, 1, 451–476. <https://doi.org/10.1007/s11292-005-3541-7>
- Lipsey, M. W., Landenberger, N. A., & Wilson, S. J. (2007). Effects of cognitive-behavioral programs for criminal offenders. *Campbell Systematic Reviews*, 3(1), 1–27. <https://doi.org/10.4073/csr.2007.6>
- Viechtbauer, W. (2022) The Metafor Package. *metafor-project*
<https://www.metafor-project.org/doku.php/features>
- Hong, C., Salanti, G., et al (2020). Testing small study effects in multivariate meta-analysis. *Biometrics*, 1, 1-2.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7736122>.

Appendix (1)

Package / Data Loading and Exploration

```
library(meta)
library(metadat)
data("dat.landemberger2005")
cbt = dat.landemberger2005
samplesizes = cbt[,8]+cbt[,9]+cbt[,10]+cbt[,11]
hist(samplesizes)
summary(samplesizes)
var(samplesizes)
```

Meta-Analysis Procedures

```
library(metafor)
eff.size.dat = escalc(measure="OR", ai=n.cbt.non, bi=n.cbt.rec, ci=n.ctrl.non,
                      di=n.ctrl.rec, data=cbt)
fixed_analysis = rma(yi,vi,method='EE',data=eff.size.dat)
summary(fixed_analysis)
random_analysis = rma(yi,vi,method='DL',data=eff.size.dat)
summary(random_analysis)
pnorm(2.68764,lower.tail=F)
forest(random_analysis)
```

Appendix (2)

```
# Meta-Regression
age = cbt[,14]; male = cbt[,15]; totalhrs = cbt[,20]; anger = cbt[,25]
for (i in 1:length(anger)) { # reformatting anger management variable
  if (anger[i]=='yes') {anger[i] = 1}
  if (anger[i]=='no') {anger[i] = 0}}
covars = cbind(age,male,numsess,totalhrs,anger)
covars = apply(covars,2,as.numeric)
cat('All Moderators \n')
model1 = rma(yi,vi,mods=~covars,data=eff.size.dat)
summary(model1)
cat('Age Only \n')
model2 = rma(yi,vi,mods=~covars[,1],data=eff.size.dat)
summary(model2)
cat('Male Proportion Only \n')
model3 = rma(yi,vi,mods=~covars[,2],data=eff.size.dat)
summary(model3)
cat('Total Hours of Therapy Only \n')
model4 = rma(yi,vi,mods=~covars[,3],data=eff.size.dat)
summary(model4)
cat('CBT Anger Control Component Only \n')
model5= rma(yi,vi,mods=~covars[,4],data=eff.size.dat)
summary(model5)
```

Appendix (3)

```
# Publication Bias
funnel(random_analysis)
colors = c("gray75", "orange", "yellow")
analysis = metagen(yi,vi,data=eff.size.dat,random=T,method.tau="DL")
funnel.meta(analysis, contour = c(0.90, 0.95, 0.99),col.contour = colors, xlab='log odds ratio')
legend(x = -1.6, y = 0.01, legend = c("p < 0.10", "p < 0.05", "p < 0.01"), fill = colors)
title("Contour-Enhanced Funnel Plot")
```

```
y = yi/vi # set up variables for quantifiable test
x = 1/vi
eggertest = lm(y~x,data=data) # egger's test
summary(eggertest) # B0 significant
plot(x,y)
xvec = seq(0,0.008,by=0.0001)
betas = eggertest$coefficients
lines(xvec,betas[1]+betas[2]*xvec,col='blue') # superimpose fitted line
title('Fitted Regression Line')
```

```
tf = trimfill(random_analysis)
summary(tf)
funnel(tf) # trimmed and filled plot
```

```
fsm(yi,vi,data=eff.size.dat)
```

Questions